

Two hours, on your own

2105623 Optimization of Chemical Processes · Semester 1 / 2026 · Monday 10 August 2026

Today's session ran for one hour instead of three. The five parts below replace the two hours we did not spend together — about two hours of work. Every file named here is on the course page:

www.skhgrou.net/teaching/2105623/

Part A Install the tools

40 minutes

Follow **2105623-software-setup.pdf** on the course page.

1. Install Python with Pyomo and the HiGHS solver.
2. Open **w01-introduction.ipynb** and run the first cell. It prints which solvers it can find.
3. In Excel, switch on the built-in **Solver** add-in. That is all you need for Week 2.

OpenSolver is optional this week. If you want it now, take **OpenSolver 2.9.4 Beta Linear** for your platform; on a Mac also run `Solvers/osx/OpenSolver Solvers.pkg` from inside the unzipped folder, then load `OpenSolver.xlam` through **Tools** → **Excel Add-ins** → **Browse**.

Keep: a screenshot of what the solver check printed. If it found no solver, say so in your submission — that is useful information, not a failure.

Part B Finish the slides

25 minutes

Open the **full** Week 1 slides on the course page — not the one-hour version — and work through the section **Anatomy, classification, geometry** to the end of the geometry slides (press **E** for a printable layout). Then answer, one or two sentences each:

- B1** Every optimization problem is exactly four things. Name them.
- B2** A solver reports *infeasible*. What does that tell you about your model, and what does it *not* tell you?
- B3** Why is rounding the answer of a linear program not a method for solving an integer program?

Part C Solve the Week 1 model

35 minutes

Download **w01-product-mix.xlsx**. The data and the layout are there; the formulas are yours to write. Fill the yellow cells, then run Solver.

- C1** The best weekly plan: how many tonnes of Additive, how many of Binder, and the total contribution margin.
- C2** Which resource constraints are *binding* at that plan — that is, fully used?
- C3** Open **w01-introduction.ipynb**, run it to the end, and confirm it gives the same objective value as your worksheet. If the two disagree, say which one you trust and why.

The point of C3. The worksheet and the Pyomo model are the same model in two dialects. Two dialects that disagree mean one of them was written wrong; finding out which is the skill this course is about.

Part D Your own problem, in five lines

15 minutes

Take a decision that you, your advisor, or your company actually has to make, and write five lines about it. This replaces the card we would have filled in class, and becomes the starting list of project topics.

- 1 The decision, in one sentence. Who makes it, and how often?
- 2 The decision variables, **with units**.
- 3 The objective. Maximum or minimum, and **with units**.
- 4 Two constraints, in words.
- 5 Your guess at the problem class: LP, NLP, IP, MILP, stochastic, or not sure.

Part E Introduce yourself

5 minutes

Four short answers, so that I know the room: your name; your research group or advisor; what you work on, in one sentence; and one thing you want to be able to do by the end of this course.

What to hand in, and when

One PDF, one or two pages, containing your answers to **B1–B3**, **C1–C3**, **D**, **E**, and the screenshot from **A**. Name the file **2105623-w01-yourID.pdf** and submit it on myCourseVille **before midnight on Friday 14 August**.

This is graded as participation, not as a test. Answers that are wrong but honestly reasoned are worth more to me than answers copied from a classmate — Week 2 opens with the two things most of you got wrong, and that only works if the answers are yours.

Stuck on Part A? Do not spend the whole two hours on it. Write down the error you got, move on to Parts B to E, and we will fix the installation at the start of Week 2.